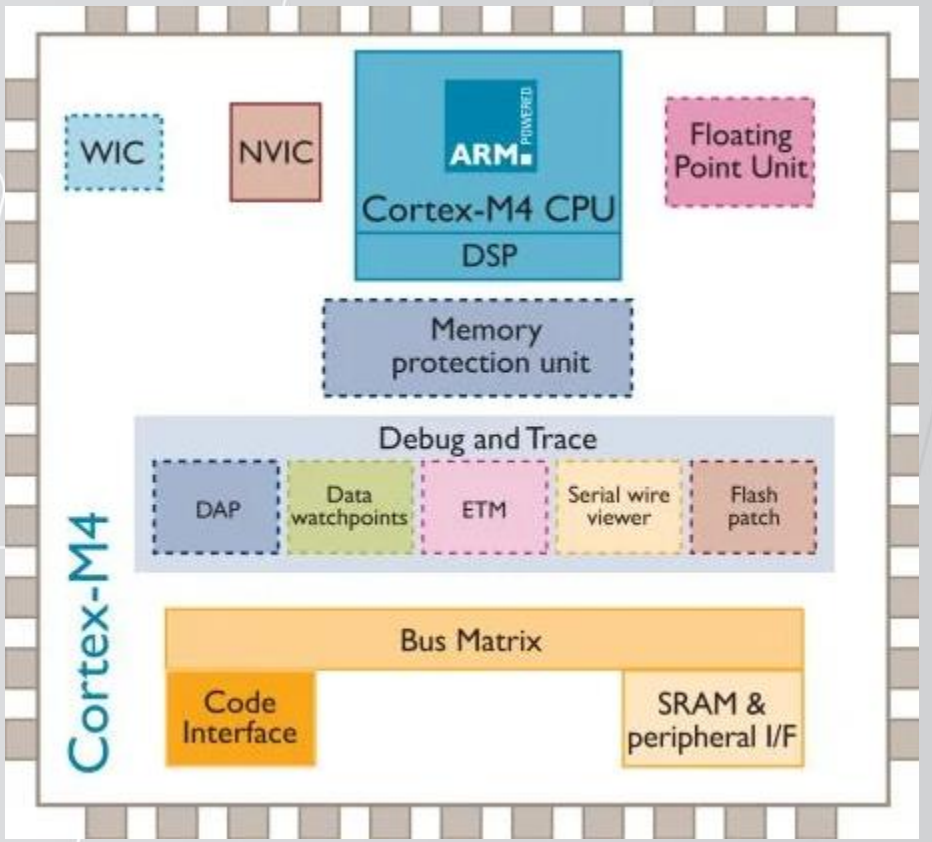
ARM Architecture



26 Oct 2022

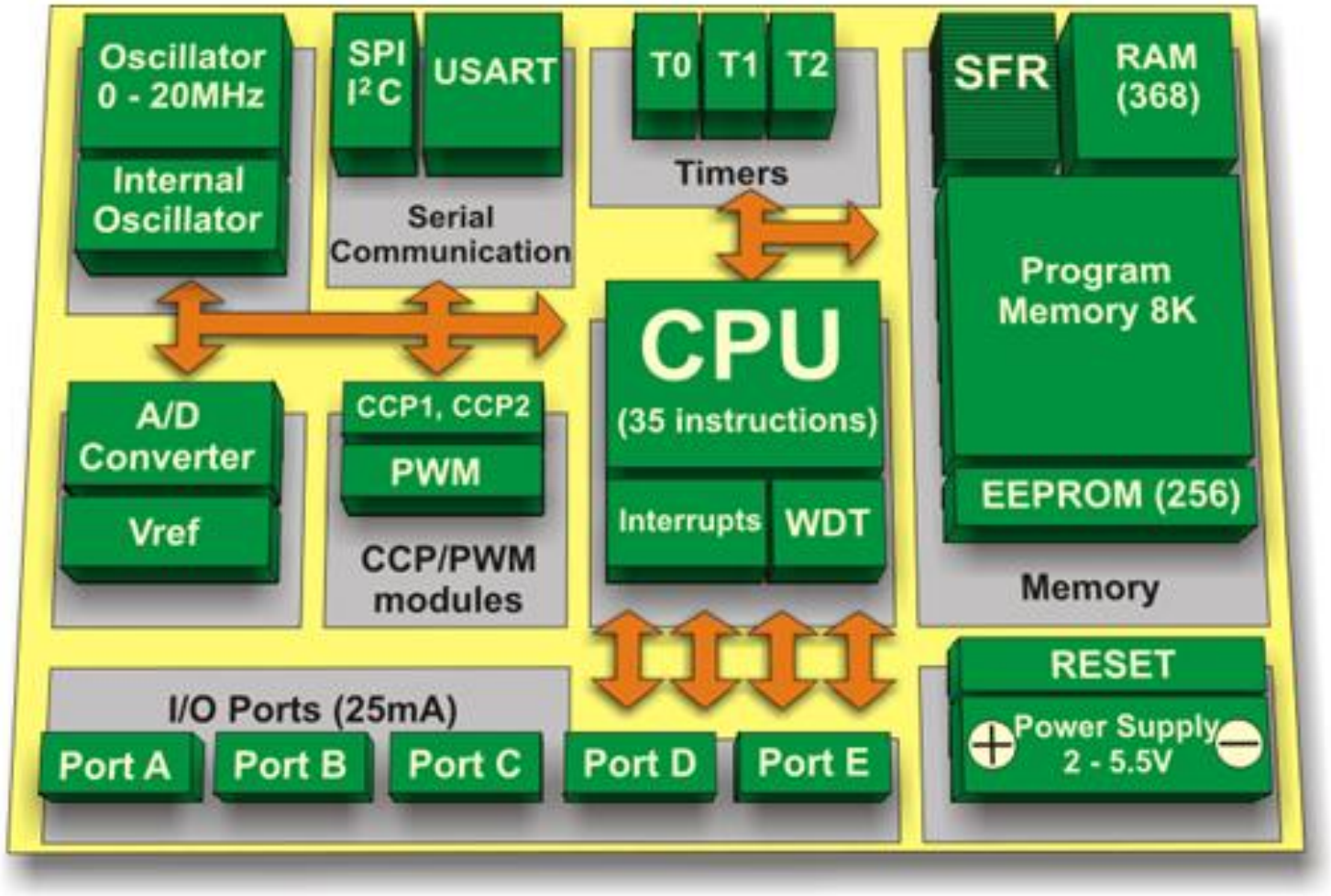
OUTLINE

1. Microcontroller
2. ARM M3 vs ARM M4 vs ARM M4F
3. ARM M7
4. ARM-M4 Block diagram
5. Cortex-M4 - Processor
6. Cortex-M4 - Floating Point Unit (FPU)
7. Cortex-M4 – NVIC
8. Memory Protection Unit (MPU)
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10. Bus interfaces
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16. Cortex-M4 – Programmers Model - Bit-banding
17. Cortex-M4 – Processor core register summary
18. Cortex-M4 – Exceptions

1. Microcontroller

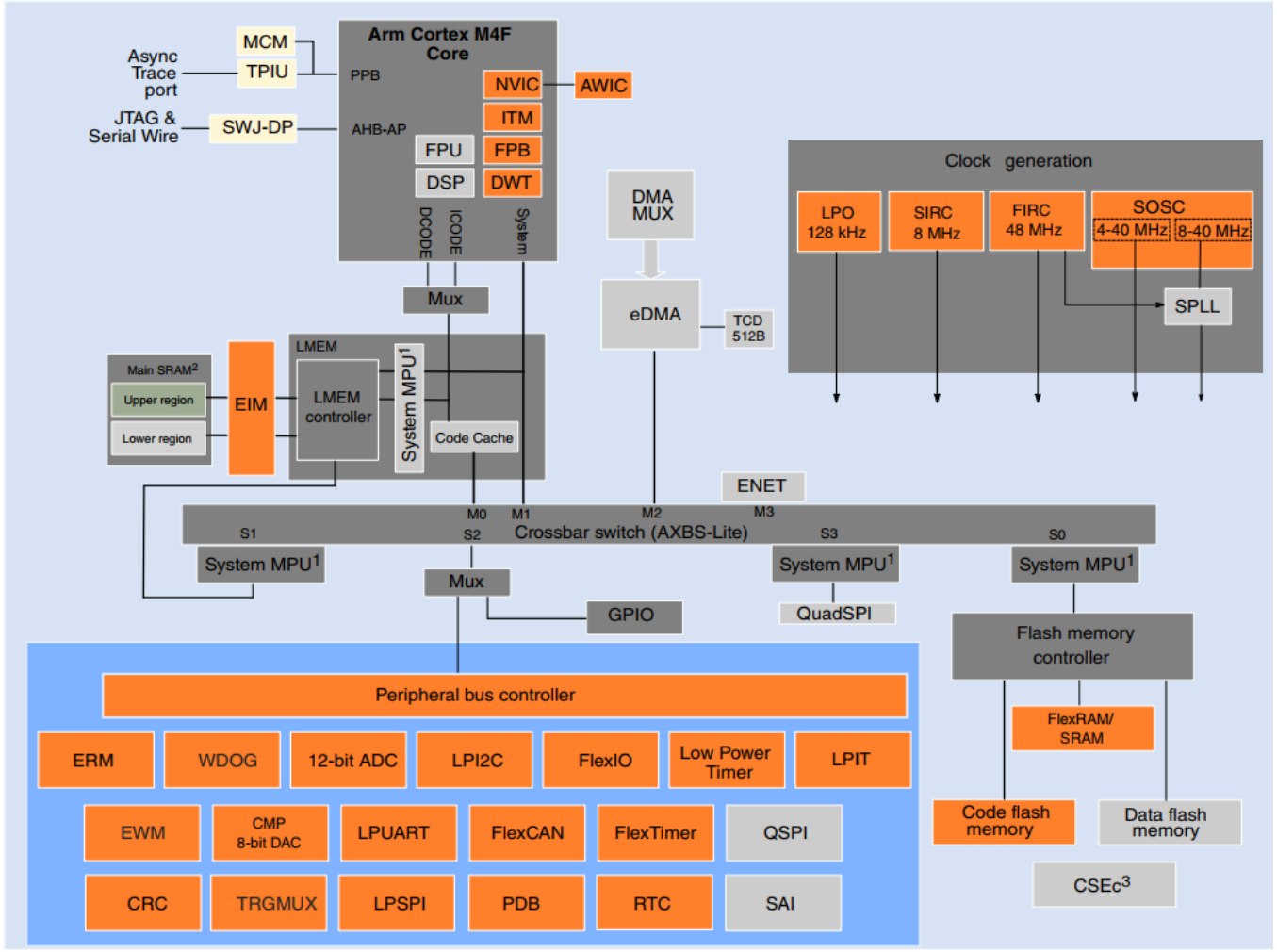
❖ Structure of Microcontroller:

+ CPU
+ Memory
+ Peripheral(PORT, TIMER,…)



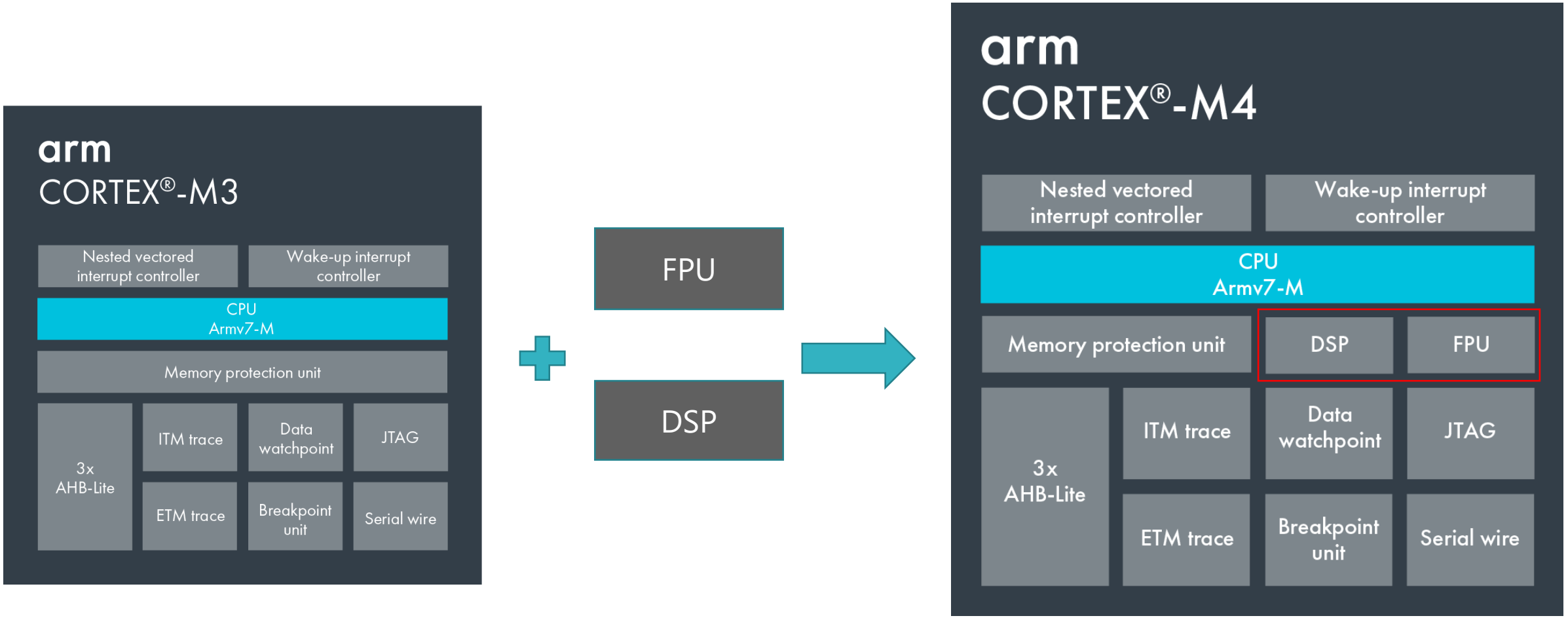
1. Microcontroller

❖ S32K14x and S32K14xW product series **block diagram**

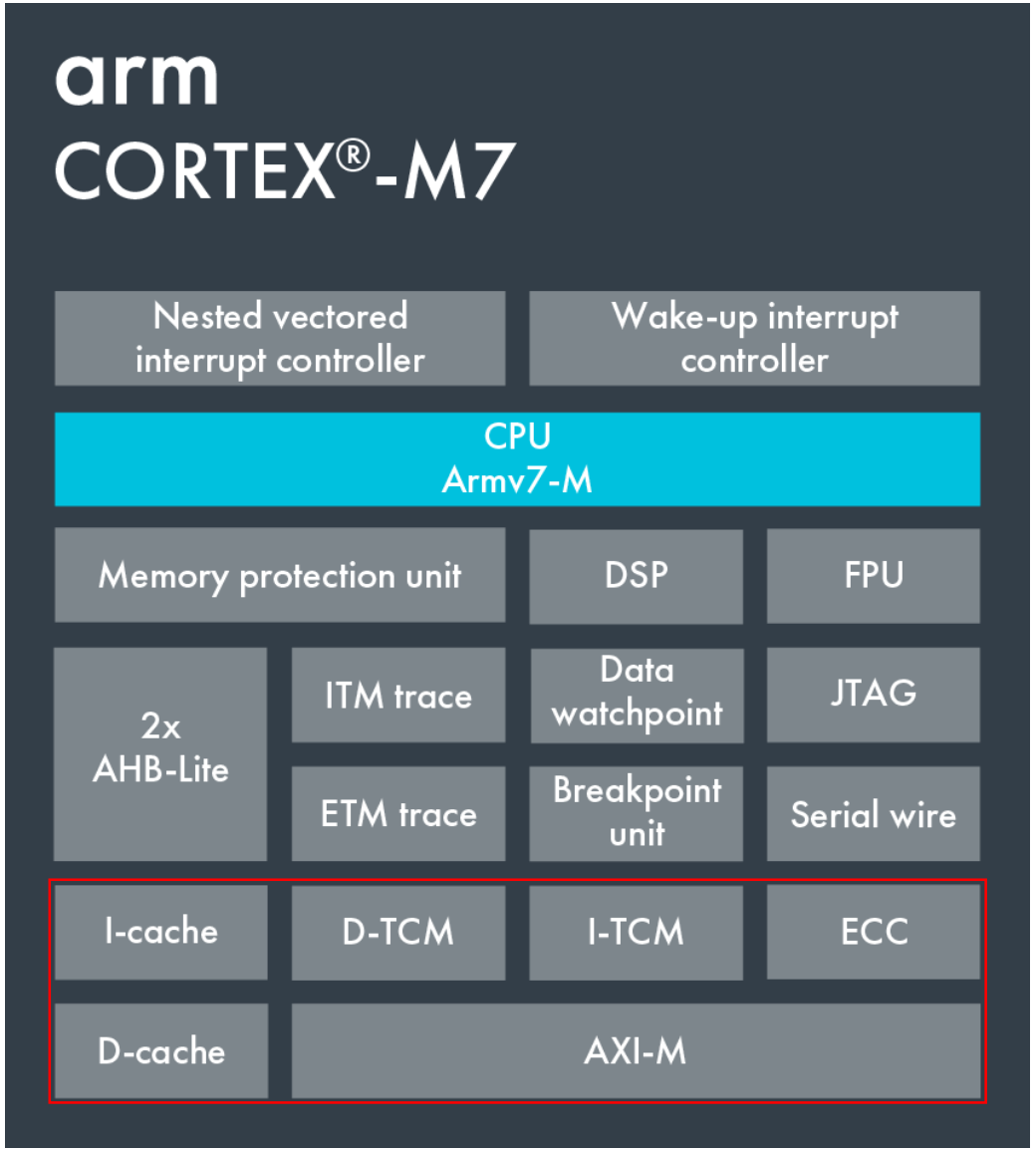


2. ARM M3 vs ARM M4 vs ARM M4F

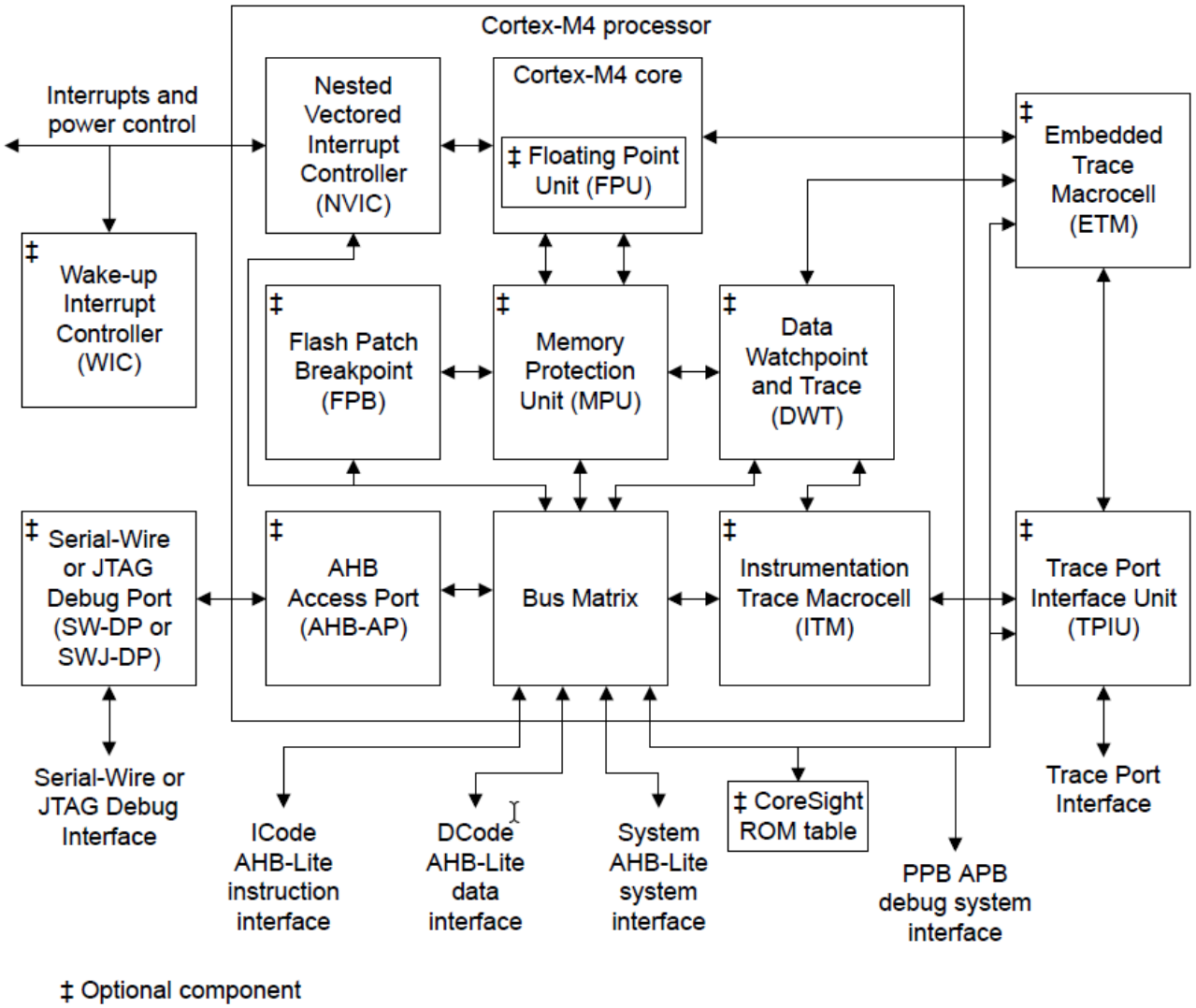
Cortex-M4 is a Cortex-M3 plus DSP instructions, and optional floating-point unit (FPU). A core with an FPU is known as Cortex-M4F.



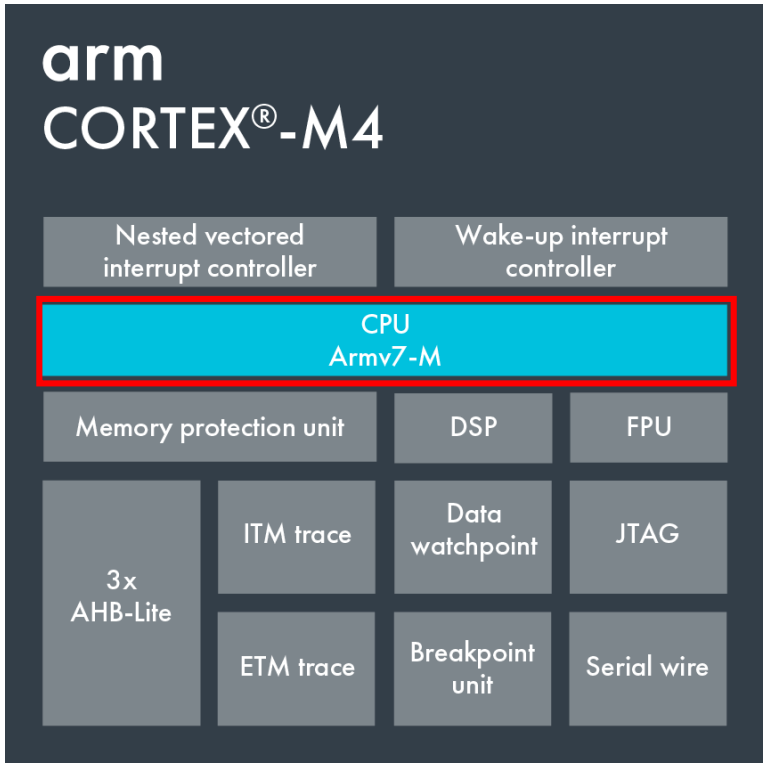
3. ARM M7


arm
CORTEX®-M7
Nested vectored interrupt controller
Wake-up interrupt controller
CPU
Armv7-M
Memory protection unit
DSP
FPU
2x AHB-Lite
ITM trace
Data watchpoint
JTAG
ETM trace
Breakpoint unit
Serial wire
I-cache
D-TCM
I-TCM
ECC
D-cache
AXI-M

4. ARM-M4 Block diagram



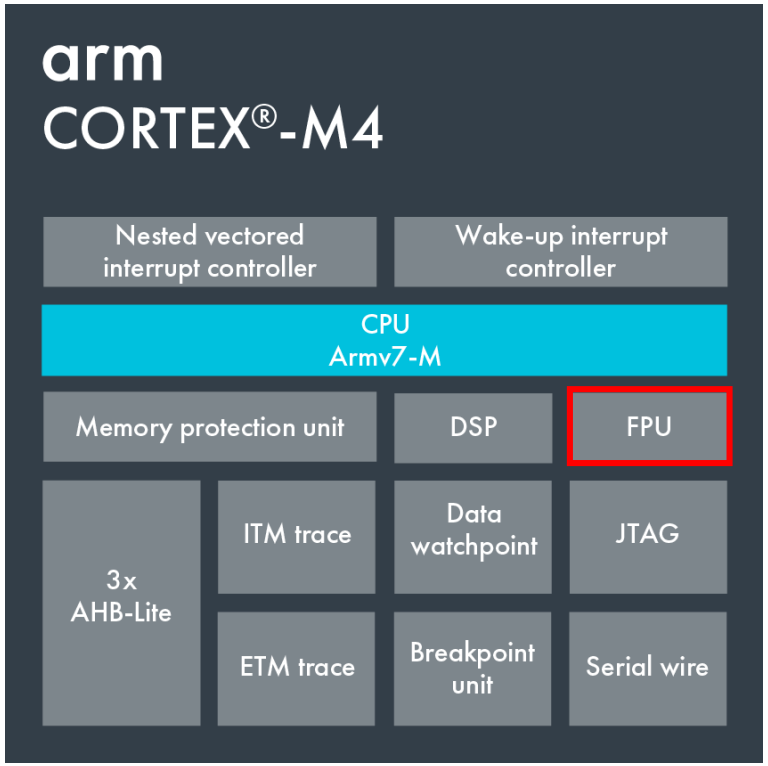
5. Cortex-M4 - Processor



The Cortex-M4 processor features a **low gate count processor** core, with **low latency interrupt processing** that has:

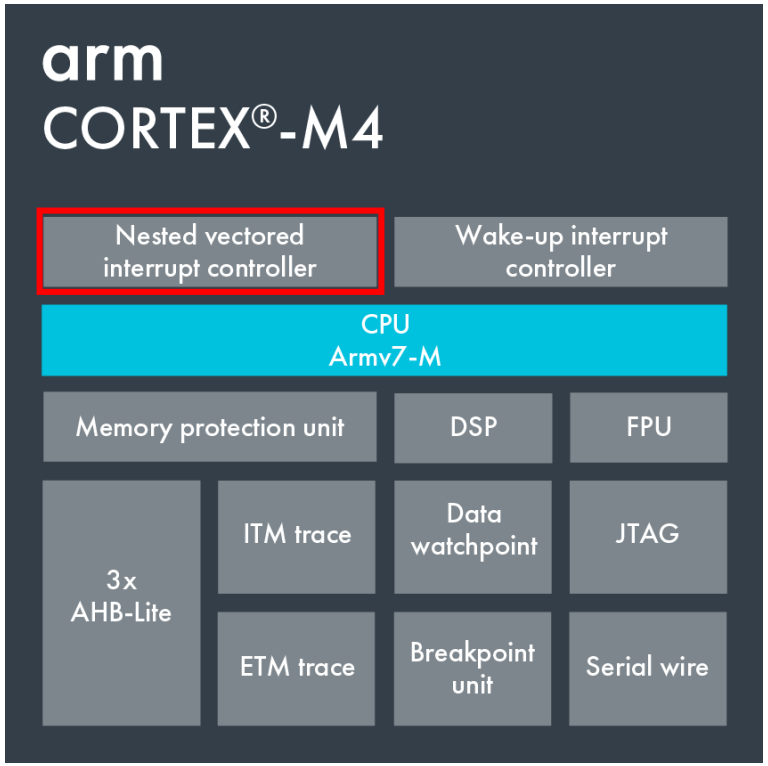
- A subset of the Thumb instruction set, defined in the Armv7-M architecture
- Banked Stack Pointer (SP)
- Hardware divide instructions, SDIV and UDIV
- Handler and Thread modes
- Thumb and Debug states
- Support for interruptible-continued instructions LDM, STM, PUSH, and POP for low interrupt latency
- Automatic processor state saving and restoration for low latency Interrupt Service
- Routine (ISR) entry and exit

6. Cortex-M4 - Floating Point Unit (FPU)



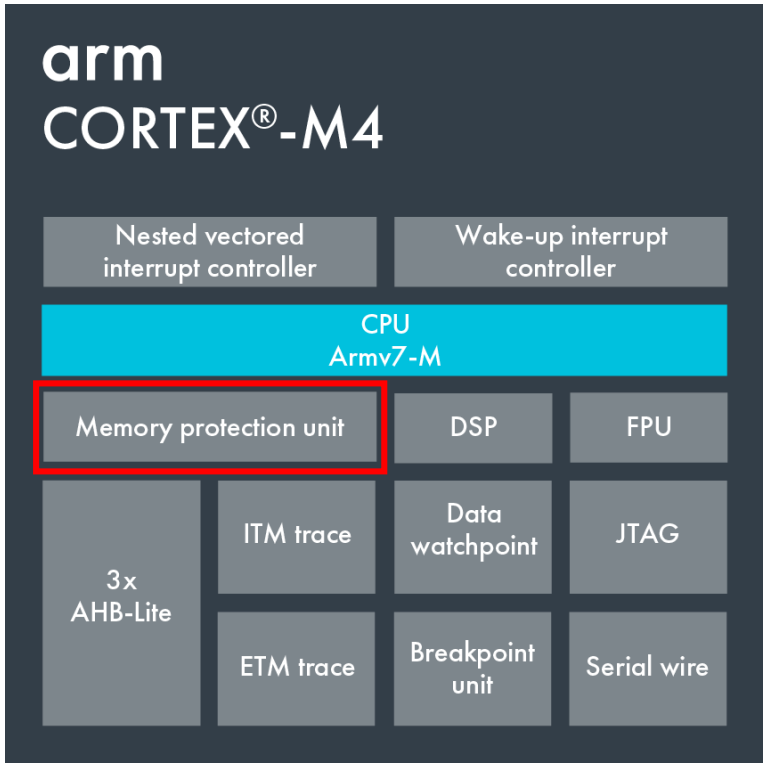
- 32-bit instructions for single-precision (C float) data-processing operations
- Combined Multiply and Accumulate instructions for increased precision (Fused MAC)
- Hardware support for conversion, addition, subtraction, multiplication with optional accumulate, division, and square root
- Hardware support for denormals and all IEEE rounding modes
- 32 dedicated 32-bit single precision registers, also addressable as 16
- Double-word registers
- Decoupled three-stage pipeline

7. Cortex-M4 – NVIC



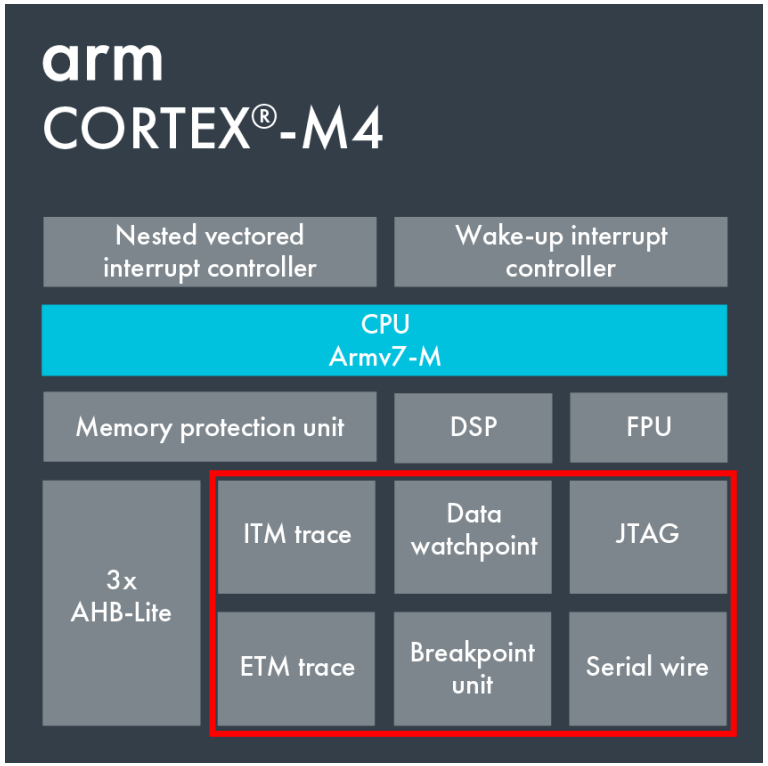
- Nested Vectored Interrupt Controller (NVIC) closely integrated with the processor core to achieve low latency interrupt processing. Features include:
  - External interrupts, configurable from 1 to 240
  - Bits of priority, configurable from 3 to 8
  - Dynamic reprioritization of interrupts
  - Priority grouping
    - This enables selection of preempting interrupt levels and non preempting interrupt levels.
  - Support for **tail-chaining** and **late arrival** of interrupts
    - This enables back-to-back interrupt processing without the overhead of state saving and restoration between interrupts.
  - Processor state automatically saved on interrupt entry, and restored on interrupt exit, with no instruction overhead
  - Optional Wake-up Interrupt Controller (WIC), providing ultra-low power sleep mode support

8. Memory Protection Unit (MPU)



- The MPU is a programmable device that can be used to define memory access permissions (e.g., privileged access only or full access) and memory attributes (e.g., bufferable, cacheable) for different memory regions.
- The MPU on CortexM3 and Cortex-M4 processors can support up to eight programmable memory regions, each with their own programmable starting addresses, sizes, and settings.
- The configuration can be used for functions like:
  - Setting a RAM/SRAM region to be read-only to protect important data from accidental corruption.
  - Making a portion of RAM/SRAM space at the bottom of the stack inaccessible to detect stack overflow.
  - Setting a RAM/SRAM region to be XN to prevent code injection attacks.

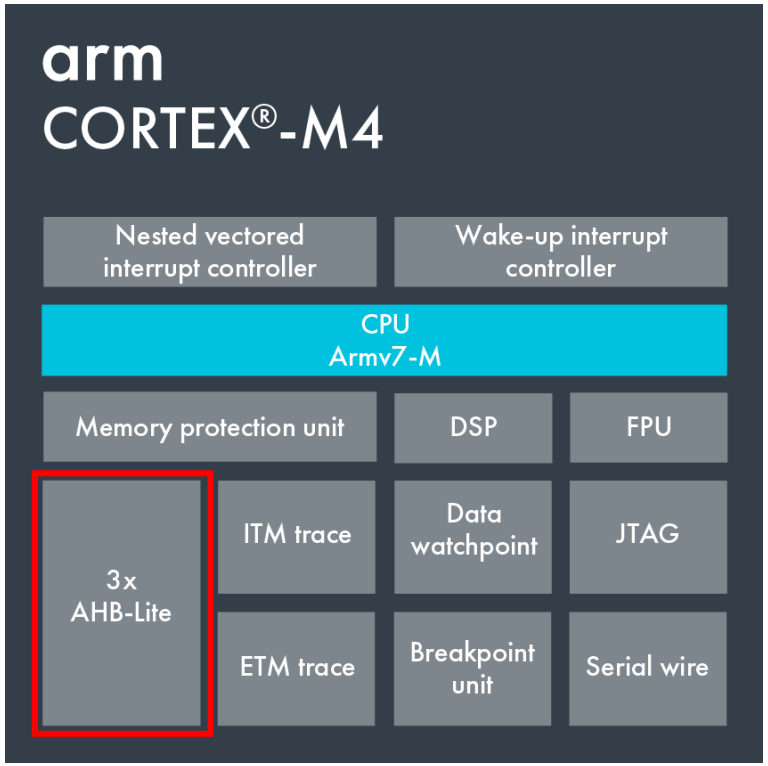
9. Debug and Trace



Low-cost debug solution that features:

- Debug access to all memory and registers in the system, including access to memory mapped devices, access to internal core registers when the core is halted, and access to debug control registers even while SYSRESETn is asserted
- Serial Wire Debug Port (SW-DP) or Serial Wire JTAG Debug Port (SWJ-DP) debug access, or both
- Optional Flash Patch and Breakpoint (FPB) unit for implementing breakpoints and code patches
- Optional Data Watchpoint and Trace (DWT) unit for implementing watchpoints, data tracing, and system profiling
- Optional Instrumentation Trace Macrocell (ITM) for support of printf style debugging
- Optional Trace Port Interface Unit (TPIU) for bridging to a Trace Port Analyzer (TPA), including Single Wire Output (SWO) mode
- Optional ETM for instruction trace

10. Bus interfaces



- Three *Advanced High-performance Bus-Lite* (AHB-Lite) interfaces: ICode, DCode, and System.
- Bus interfaces.
- *Private Peripheral Bus* (PPB) based on *Advanced Peripheral Bus* (APB) interface.
- Bit-band support that includes atomic bit-band write and read operations.
- Memory access alignment.
- Write buffer for buffering of write data.
- Exclusive access transfers for multiprocessor systems.

11. Programmers model Mode – Level – State

Processor mode

Thread mode
Handler mode

Privilege levels

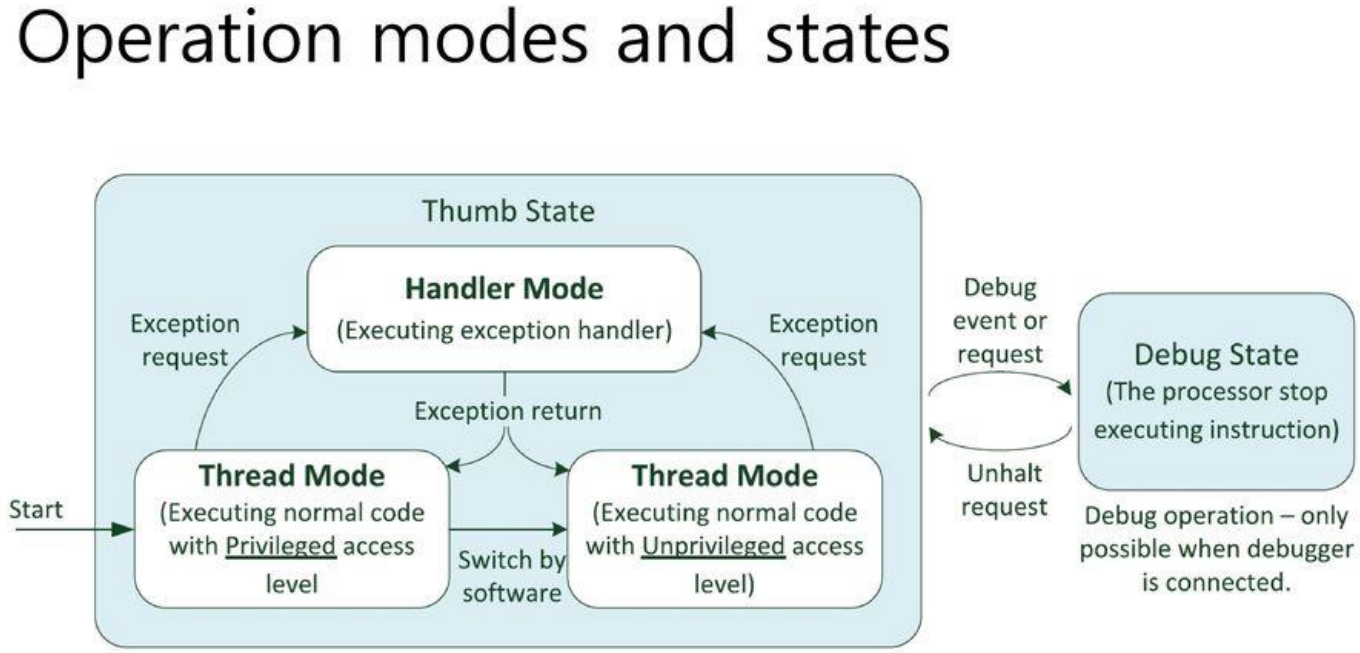
Unprivileged
Privileged

State

ARM state
Thumb State
Debug State

Arm code
Thumb code

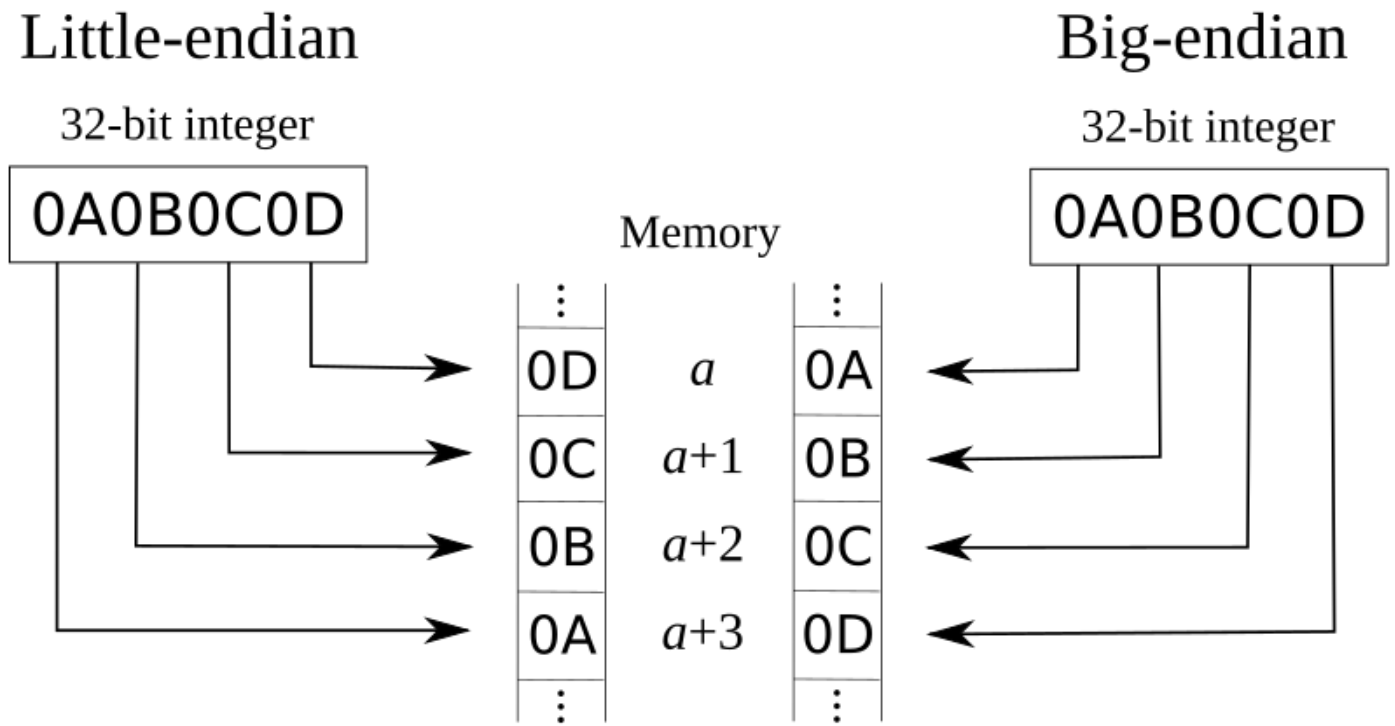
https://kmittal82.wordpress.com/2012/02/17/armthumbthumb-2/



11. Programmers model – Data Types – LE – BE

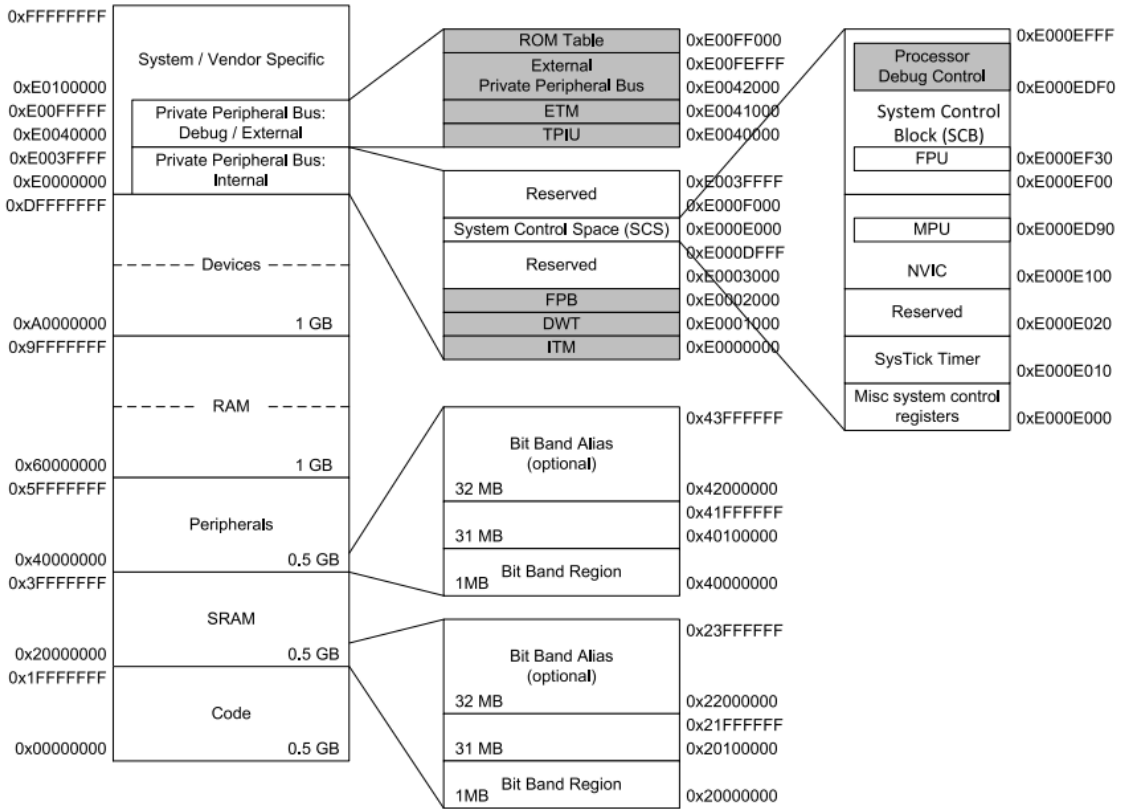
- Data types:
  - 32-bit words
  - 16-bit half-words
  - 8-bit bytes

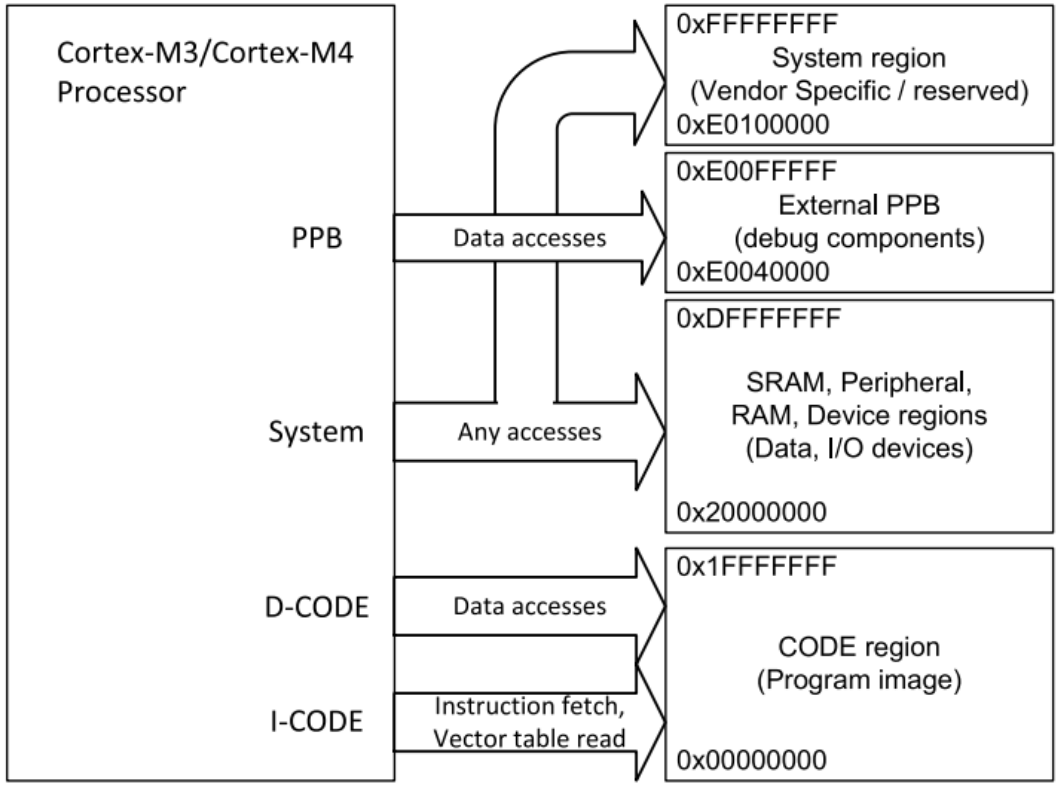
- Instruction memory and *Private Peripheral Bus* (PPB) accesses are always performed as **little-endian.**



11. Memory model – memory map – regions

❖ Up to 4GB of addressable memory

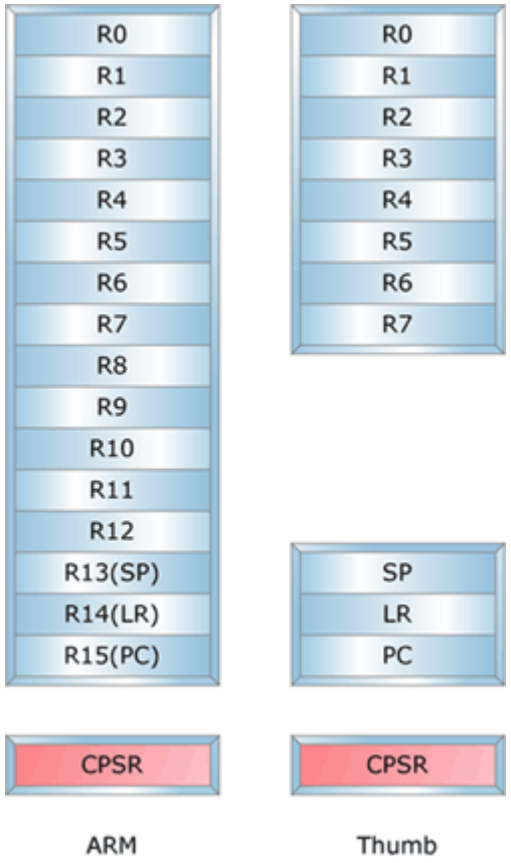




11. Cortex-M4 – ARM thumb

+ Advanced RISC Machines - (ARM)
+ CISC processor vs RISC processor
❖ RISC technology
+ execute each instruction in a single clock cycle
+ faster execution than on a CISC processor
+ RISC processor typically needs more memory than a CISC does to store the same program.

+ ARM state vs Thumb state.
Current Program Status Register (CPSR)
SPSR (Saved Program Status Register)
Stack Pointer (SP).



https://www.embedded.com/introduction-to-arm-thumb/
http://embeddedknowledge.blogspot.com/2012/03/difference-between-arm-and-thumb-states.html

12. Modes of operation and execution

Operating modes

1) Thread mode :

The processor enters Thread mode on Reset, a result of an exception return. Privileged and Unprivileged code can run in Thread mode

2) Handler mode :

The processor enters Handler mode as a result of an exception. All code is privileged in Handler mode.

Operating states

1) Thumb state. This is normal execution running 16-bit and 32-bit halfword aligned Thumb instructions.

2) Debug State. This is the state when the processor is in halting debug

Access Level :

Privileged access

Unprivileged access

User access

Handler mode is always privileged. Thread mode can be privileged or unprivileged.

13. Cortex-M4 – Bus Interface

- **ICode memory interface :** Instruction fetches from Code memory space, 0x00000000 to 0x1FFFFFFC, are performed over the 32-bit AHB-Lite bus.
- **DCode memory interface** : Data and debug accesses to Code memory space, 0x00000000 to 0x1FFFFFFF, are performed over the 32-bit AHB-Lite bus.
- **System interface :** Instruction fetches and data and debug accesses to address ranges 0x20000000 to 0xDFFFFFFF and 0xE0100000 to 0xFFFFFFFF are performed over the 32-bit AHB-Lite bus.
- **Private Peripheral Bus (PPB) :** Data and debug accesses to external PPB space, 0xE0040000 to 0xE00FFFFF, are performed over the 32-bit Advanced Peripheral Bus (APB) bus. The *Trace Port Interface Unit* (TPIU) and vendor specific peripherals are on this bus.
- **ETM interface :** The AHB Trace Macrocell (HTM) interface enables a simple connection of the AHB trace macrocell to the processor, and provides a channel for the data trace to the HTM.
- **Debug Port AHB-AP interface :** The processor contains an Advanced High-performance Bus Access Port (AHB-AP) interface for debug accesses. An external Debug Port (DP) component accesses this interface.

14. Cortex-M4 – Instruction set summary

Processor instructions

Operation
Move
Add
Subtract
Multiply
Divide
Saturate
Compare
Logical
Shift
Rotate
Count
Store
Push

Operation
Pop
Semaphore
Branch
State change
Extend
Bit field
Reverse
Hint
Barriers

15. Cortex-M4 – Processor memory model

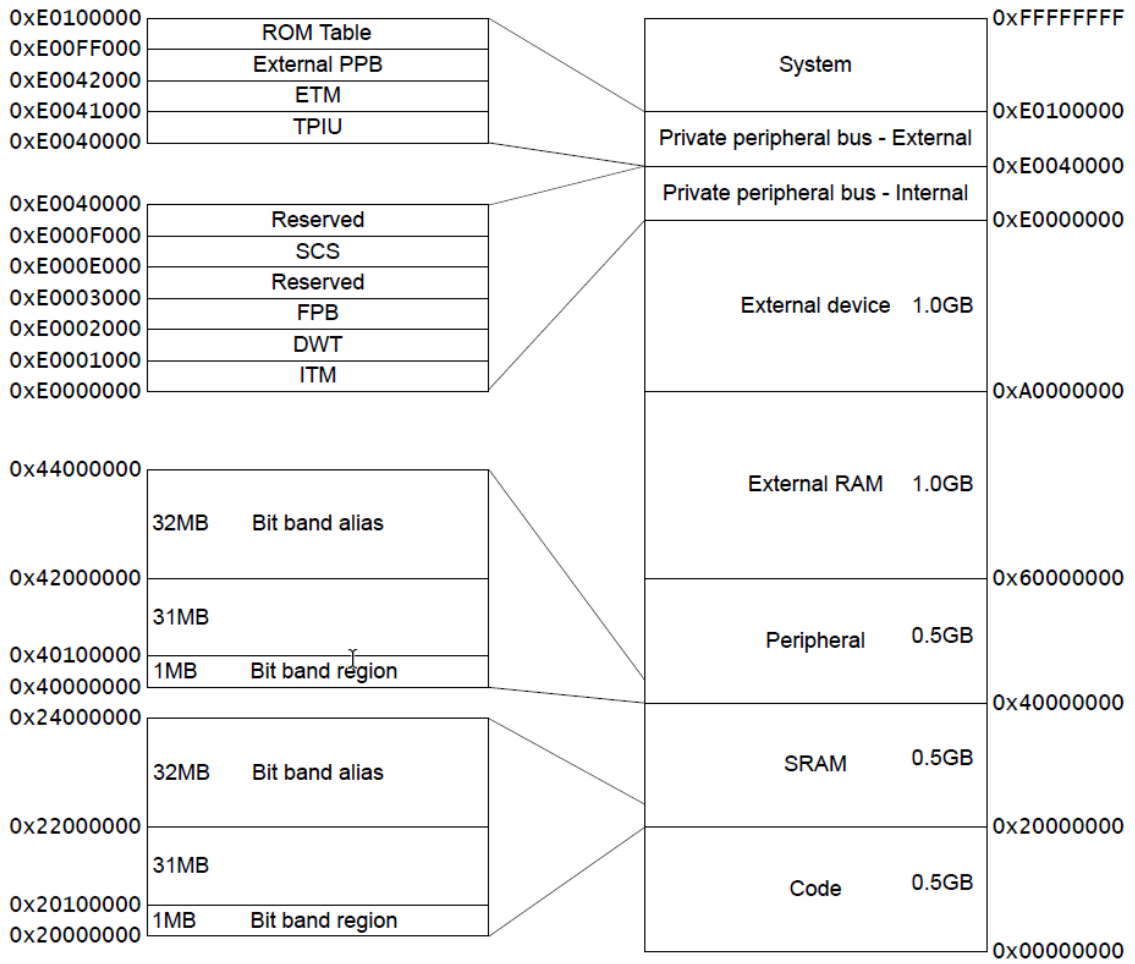
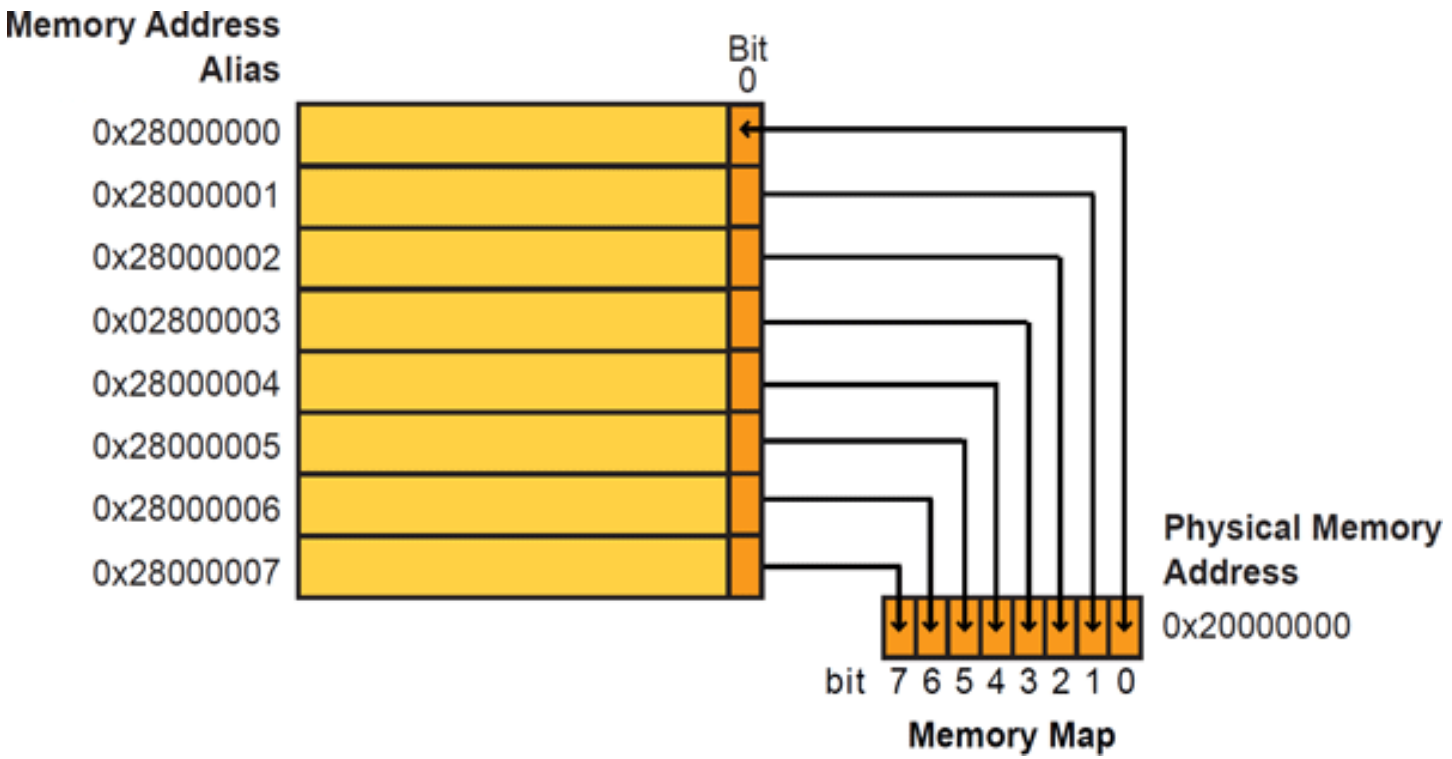


Figure 3-1 System address map

Memory Map	Region
Code	Instruction fetches are performed over the ICode bus. Data accesses are performed over the DCode bus.
SRAM	Instruction fetches and data accesses are performed over the system bus.
SRAM bit-band	Alias region. Data accesses are aliases. Instruction accesses are not aliases.
Peripheral	Instruction fetches and data accesses are performed over the system bus.
Peripheral bit-band	Alias region. Data accesses are aliases. Instruction accesses are not aliases.
External RAM	Instruction fetches and data accesses are performed over the system bus.
External Device	Instruction fetches and data accesses are performed over the system bus.
Private Peripheral Bus	External and internal *Private Peripheral Bus* (PPB) interfaces. This memory region is *Execute Never* (XN), and so instruction fetches are prohibited. An MPU, if present, cannot change this.
System	System segment for vendor system peripherals. This memory region is XN, and so instruction fetches are prohibited. An MPU, if present, cannot change this.

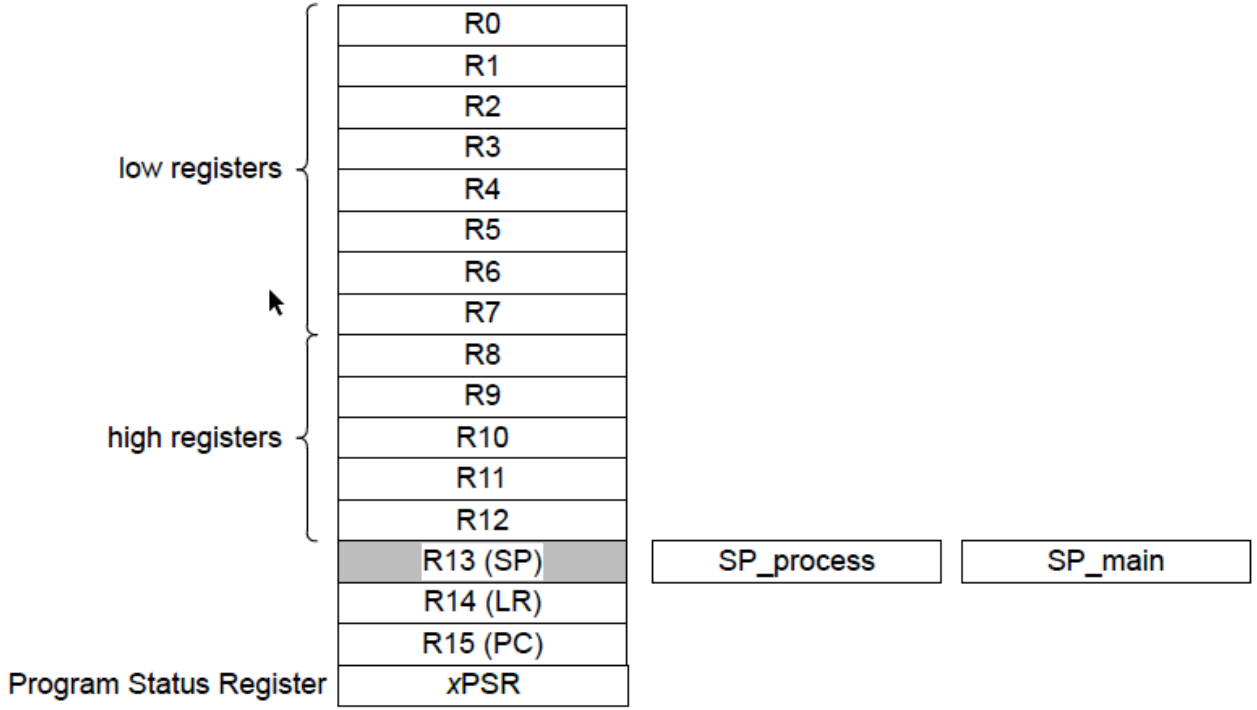
16. Cortex-M4 – Programmers Model - Bit-banding



17. Cortex-M4 – Processor core register summary

The processor has the following 32-bit registers:

- 13 general-purpose registers, R0-R12.
- *Stack Pointer* (SP), R13 alias of banked registers, SP_process and SP_main.
- *Link Register* (LR), R14.
- *Program Counter* (PC), R15.
- Special-purpose *Program Status Registers*, (*x*PSR).

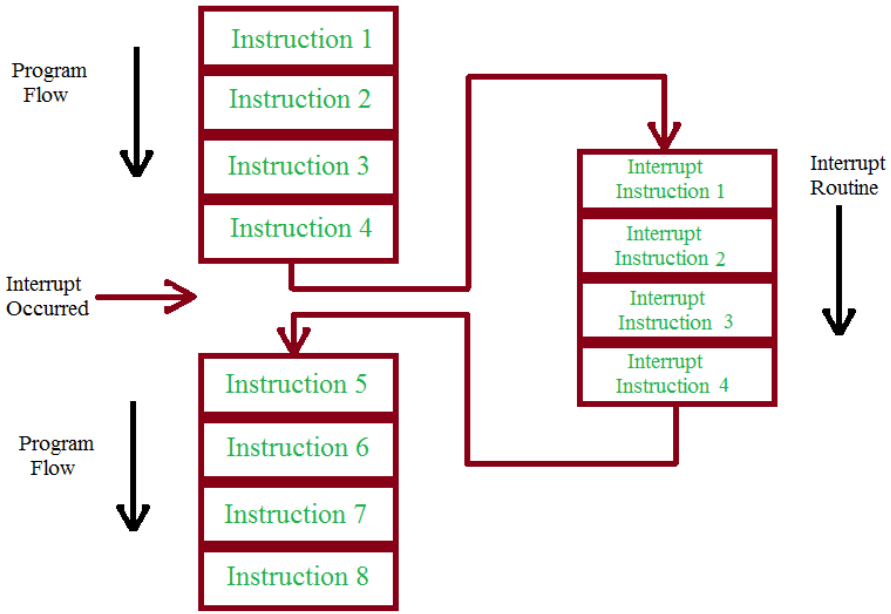


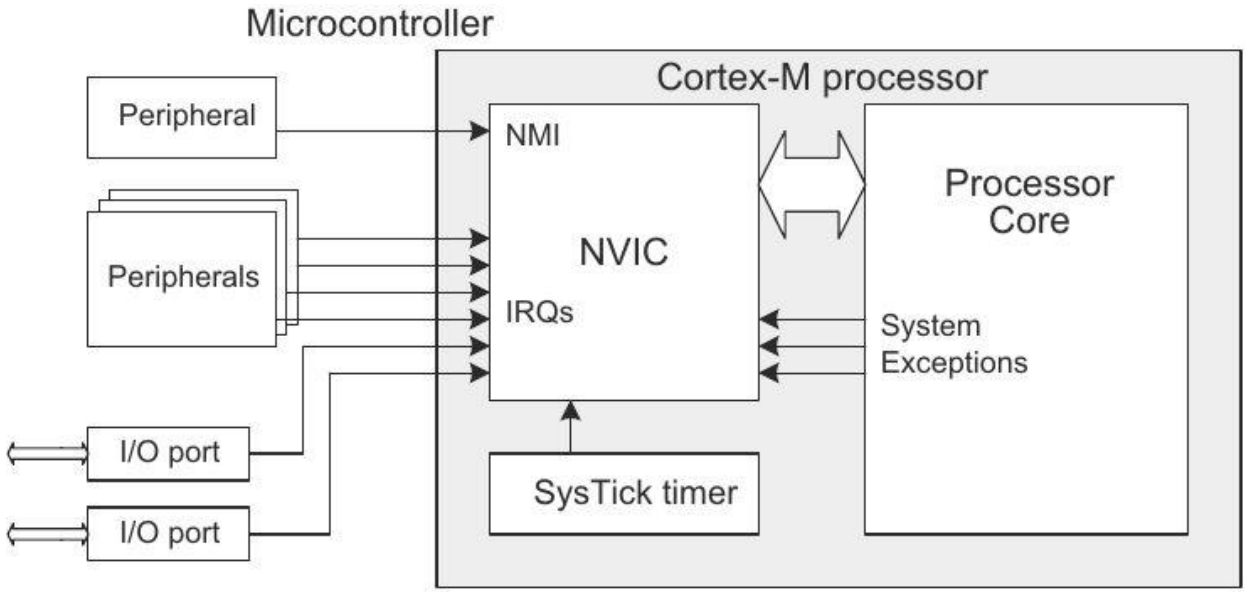
SP: Handler mode always uses SP_main, but you can configure Thread mode to use either SP_main or SP_process.

LR: The LR receives the return address from PC

18. Cortex-M4 – Exceptions

Exceptions are handled and prioritized by the processor and the NVIC. All exceptions are handled in Handler mode.





Technical documents

1. Cortex-M4 Technical Reference Manual - TRM
2. Cortex-M4 Integration and Implementation Manual – available as part of the Bill of Materials
3. Armv7-M Architecture Reference Manual - ARM
4. CoreSight ETM-M4 Technical Reference Manual – ETM
5. Cortex-M4-Processor-Datasheet


Thank you